Approximating Values using the Binomial Expansion

Exam code: 9MA0

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Approximating values using the binomial expansion

What is the general binomial expansion?

- The **general binomial expansion**, as given in the formula booklet, is

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1\times 2}x^2 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{1\times 2\times\ldots\times r}x^r + \ldots \qquad (|x|<1,\ n\in\mathbb{R})$$

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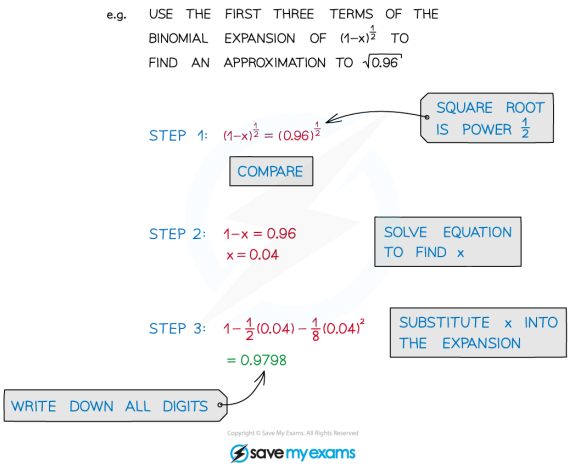
- If $n \in \mathbb{N}$ then the expansion is **finite** (see Binomial Expansion)
- Otherwise the expansion is infinitely long
  - It is only valid for $|x| < 1$ ($-1 < x < 1$)
  - Only the first few terms of an expansion are usually needed

How do I use a binomial expansion to approximate a value?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Ignoring higher powers of **x** leads to an approximation
- The more terms the closer the approximation is to the true value
- For most purposes, squared or cubed terms are accurate enough



STEP 1 Compare the value you are approximating to $(\mathbf{a} + \mathbf{bx})^n$
STEP 2 Solve the appropriate equation to find the value of **x**
STEP 3 Substitute this value of **x** into the expansion to find the approximation

Examiner Tips and Tricks

- You can get a good idea if your approximation is correct by working out the "real" answer using your calculator.
- Sometimes it helps to factorise out a number before approximating
  - $\sqrt{710} = 10\sqrt{7.1}$

Worked Example

expansion of $\sqrt[4]{81-9x}$

(b) Use your expansion from part (a) to approximate $\sqrt[4]{85}$

(c) Explain why your expansion would not be suitable for approximating $\sqrt[4]{171}$

a)

$$\sqrt[4]{81-9x} = (81-9x)^{\frac{1}{4}}$$
$$= 81^{\frac{1}{4}}\left(1-\frac{1}{9}x\right)^{\frac{1}{4}}$$
$$= 3\left[1+\left(\frac{1}{4}\right)\left(-\frac{1}{9}x\right) + \frac{\left(\frac{1}{4}\right)\left(-\frac{3}{4}\right)}{2}\left(-\frac{1}{9}x\right)^2 + \ldots\right]$$
$$\approx 3\left(1-\frac{1}{36}x-\frac{1}{864}x^2\right)$$

$$\therefore \sqrt[4]{81-9x} \approx 3-\frac{1}{12}x-\frac{1}{288}x^2$$

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b)

STEP 1: $\sqrt[4]{81-9x} = \sqrt[4]{85}$ COMPARE

STEP 2: $81-9x = 85$

$9x = -4$

$x = -\frac{4}{9}$ SOLVE TO FIND x

STEP 3: $\sqrt[4]{81-9x} \approx 3-\frac{1}{12}\left(-\frac{4}{9}\right)-\frac{1}{288}\left(-\frac{4}{9}\right)^2$

$=3.036351166$ SUBSTITUTE INTO EXPANSION

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